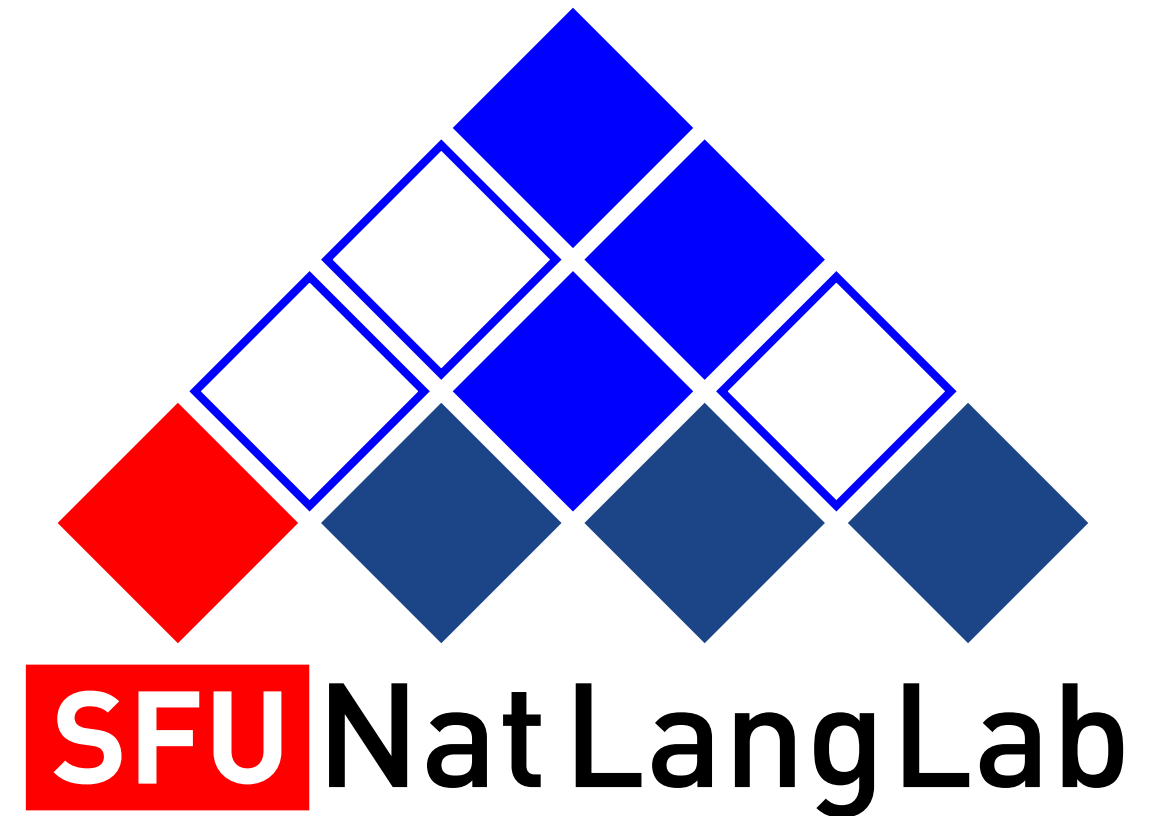




CMPT 413/713: Natural Language Processing

Text Classification Evaluation

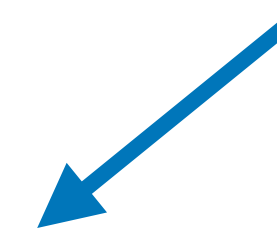
Fall 2021
2021-09-16

Adapted from slides from Anoop Sarkar, Danqi Chen and Karthik Narasimhan

Evaluation

- Consider binary classification
- Table of predictions

Confusion matrix



		<i>Truth</i>	
		Positive	Negative
<i>Predicted</i>	Positive	100	5
	Negative	45	100

- Ideally, we want:

		<i>Truth</i>	
		Positive	Negative
<i>Predicted</i>	Positive	145	0
	Negative	0	105

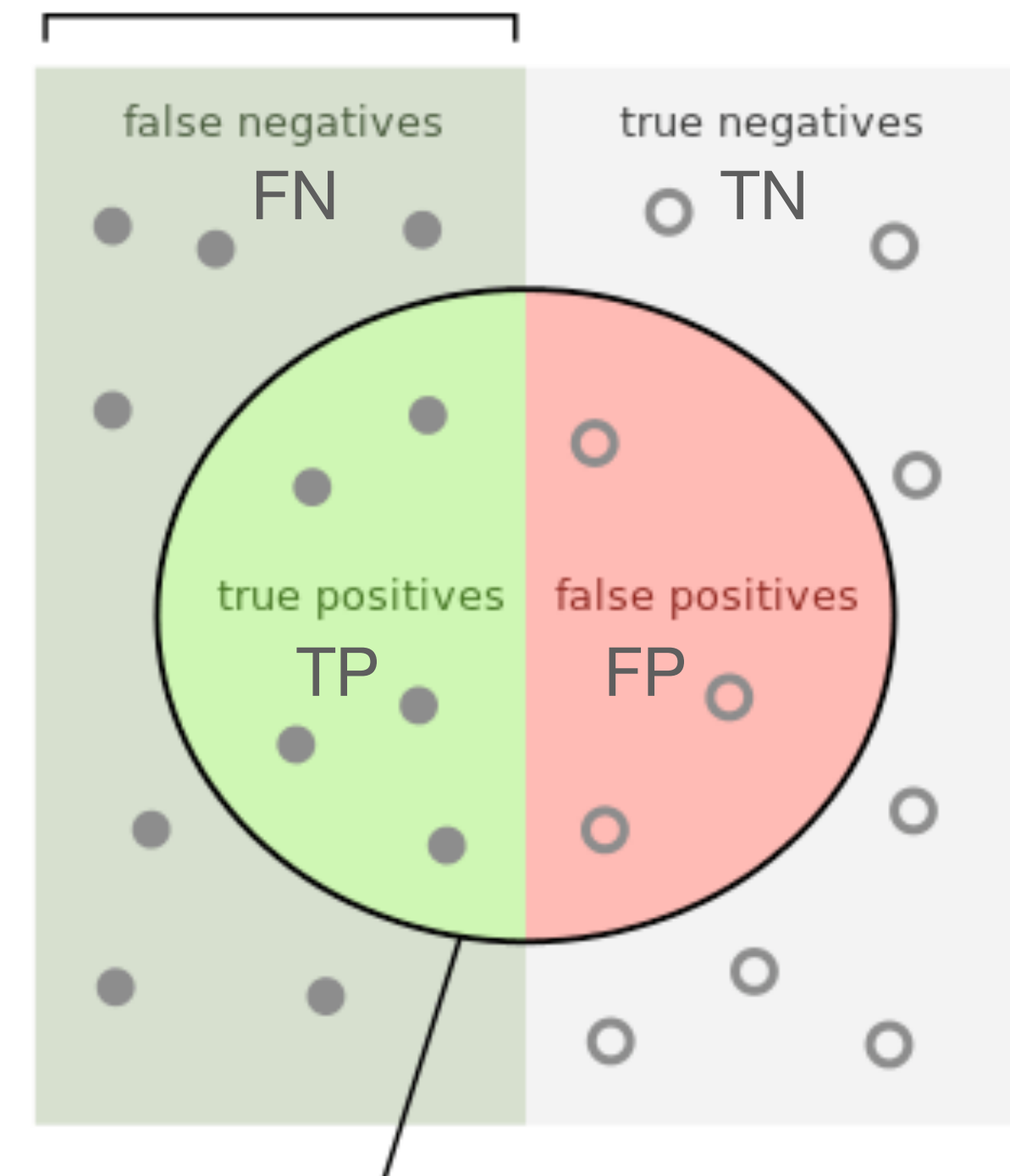
Evaluation Metrics

		Truth	
		Positive	Negative
Predicted	Positive	100 TP	5 FP
	Negative	45 FN	100 TN

- True positive (TP): Predicted + and actual +
- True negative (TN): Predicted - and actual -
- False positive (FP): Predicted + and actual -
- False negative (FN): Predicted - and actual +

$$\text{Accuracy} = \frac{TP + TN}{\text{Total}} = \frac{200}{250} = 80\%$$

Actual positives



Predicted positives

(image credit: wikipedia)

Evaluation Metrics

</

- True positive (TP): Predicted + and actual +
- True negative (TN): Predicted - and actual -
- False positive (FP): Predicted + and actual -
- False negative (FN): Predicted - and actual +

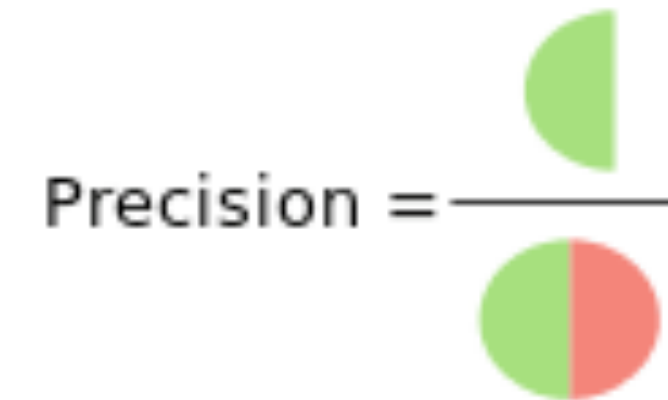
Coarse metric

$$\text{Accuracy} = \frac{TP + TN}{Total} = \frac{200}{250} = 80 \%$$

Precision and Recall

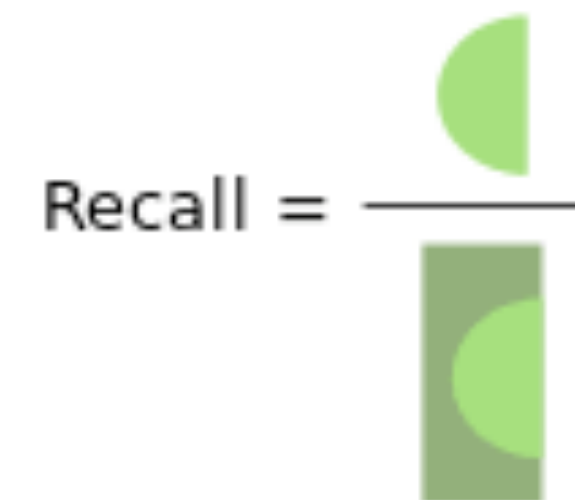
- Precision: % of selected classes that are correct

$$\text{Precision}(+) = \frac{TP}{TP + FP}$$

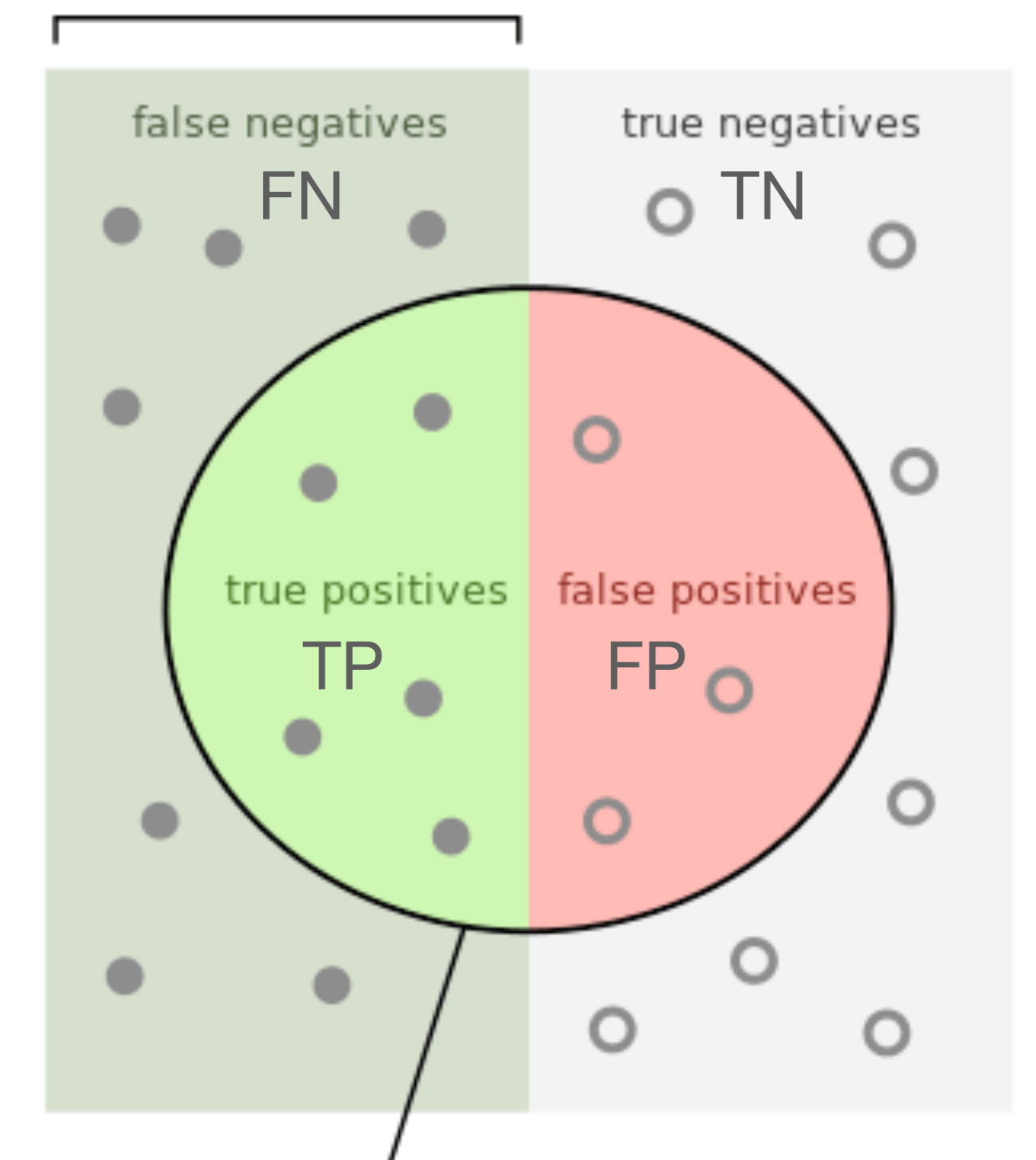


- Recall: % of correct items selected

$$\text{Recall}(+) = \frac{TP}{TP + FN}$$



Actual positives (relevant)



Predicted positives
(selected/retrieved)

(image credit: wikipedia)

Precision and Recall

- Precision: % of selected classes that are correct

$$\text{Precision}(+) = \frac{TP}{TP + FP}$$

$$\text{Precision}(-) = \frac{TN}{TN + FN}$$

- Recall: % of correct items selected

$$\text{Recall}(+) = \frac{TP}{TP + FN}$$

$$\text{Recall}(-) = \frac{TN}{TN + FP}$$

Evaluation Metrics

		Truth	
		Positive	Negative
Predicted	Positive	100	5
	Negative	45	100

		Truth	
		Positive	Negative
Predicted	Positive	50	25
	Negative	25	150

$$\text{Precision}(+) = \frac{TP}{TP + FP}$$

$$\frac{100}{100 + 5} = 0.95$$

$$\frac{50}{50 + 25} = 0.75$$

$$\text{Recall}(+) = \frac{TP}{TP + FN}$$

$$\frac{100}{100 + 45} = 0.69$$

$$\frac{50}{50 + 25} = 0.75$$

Two metrics - which one to use?

F-Score

- Combined measure
- Harmonic mean of Precision and Recall

$$F_1 = \frac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$$

- Or more generally,

$$F_\beta = \frac{(1 + \beta^2) \cdot \text{Precision} \cdot \text{Recall}}{\beta^2 \cdot \text{Precision} + \text{Recall}}$$

Evaluation Metrics

		Truth	
		Positive	Negative
Predicted	Positive	100	5
	Negative	45	100

		Truth	
		Positive	Negative
Predicted	Positive	50	25
	Negative	25	150

$$\text{Precision}(+) = \frac{TP}{TP + FP}$$

$$\frac{100}{100 + 5} = 0.95$$

$$\frac{50}{50 + 25} = 0.75$$

$$\text{Recall}(+) = \frac{TP}{TP + FN}$$

$$\frac{100}{100 + 45} = 0.69$$

$$\frac{50}{50 + 25} = 0.75$$

$$F_1(+) = \frac{2 \cdot P(+)R(+)}{P(+) + R(+)}$$

$$0.8$$

$$0.75$$

Evaluation Metrics

Q: What happens to $F_1(+)$ if $FN = 25$ and $FP = 25$?

		Truth	
		Positive	Negative
Predicted	Positive	100	5
	Negative	45	100

	Positive	Negative
Positive	100	25
Negative	25	100

$$\text{Precision}(+) = \frac{TP}{TP + FP} = \frac{100}{100 + 5} = 0.95$$

$$\text{Recall}(+) = \frac{TP}{TP + FN} = \frac{100}{100 + 45} = 0.69$$

$$F_1(+) = \frac{2 \cdot P(+)R(+)}{P(+) + R(+)} = 0.8$$

Evaluation Metrics

		<i>Truth</i>	
		Positive	Negative
<i>Predicted</i>	Positive	TP	FP
	Negative	FN	TN

Use a simple rule, can you design a classifier with

$$\text{Precision}(+) = \frac{TP}{TP + FP}$$

Q. perfect precision?

$$\text{Recall}(+) = \frac{TP}{TP + FN}$$

Q. perfect recall?

Choosing Beta

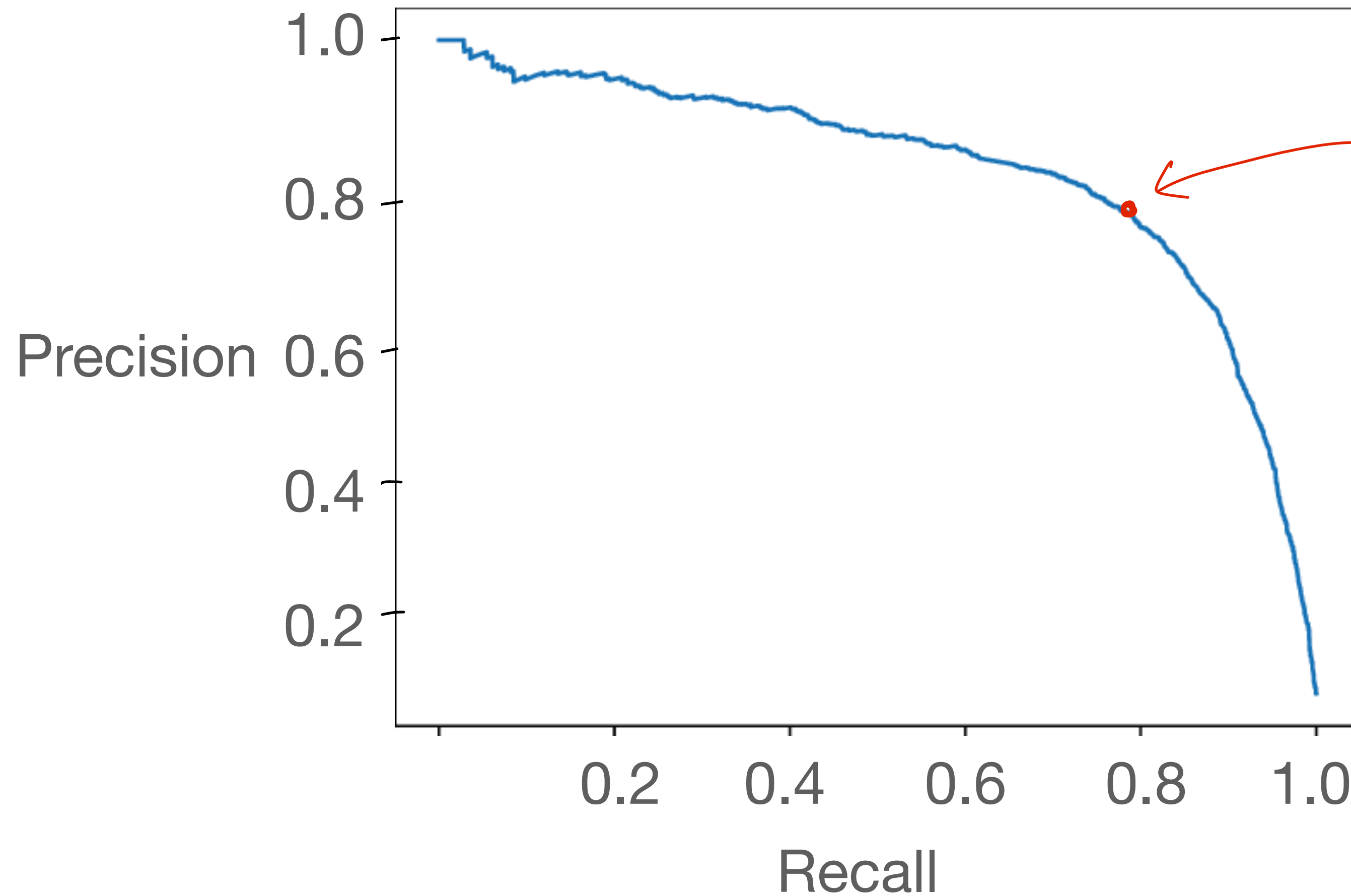
		<i>Truth</i>	
		Positive	Negative
<i>Predicted</i>	Positive	200	100
	Negative	50	100

$$F_{\beta} = \frac{(1 + \beta^2) \cdot \text{Precision} \cdot \text{Recall}}{\beta^2 \cdot \text{Precision} + \text{Recall}}$$

Q. Which value of Beta maximizes F_{β} for positive class?

- A. $\beta = 0.5$
- B. $\beta = 1$
- C. $\beta = 2$

Precision Recall tradeoff



Maximum F1

Vary

- Smoothing α
- Threshold T

$$\frac{P(+ | d)}{P(- | d)} > T$$

Tune on validation set

Train, val, test split

- Train model on **training** set
- Tune hyperparameters on **validation** set
- Evaluate performance on unseen **test** set



Why do we do this?

More training examples,
Less “accidental regularities”,
Generalization (test) error goes down



Smaller dataset
are easier to
memorize

Smaller network
has less capacity
to memorize the
training data

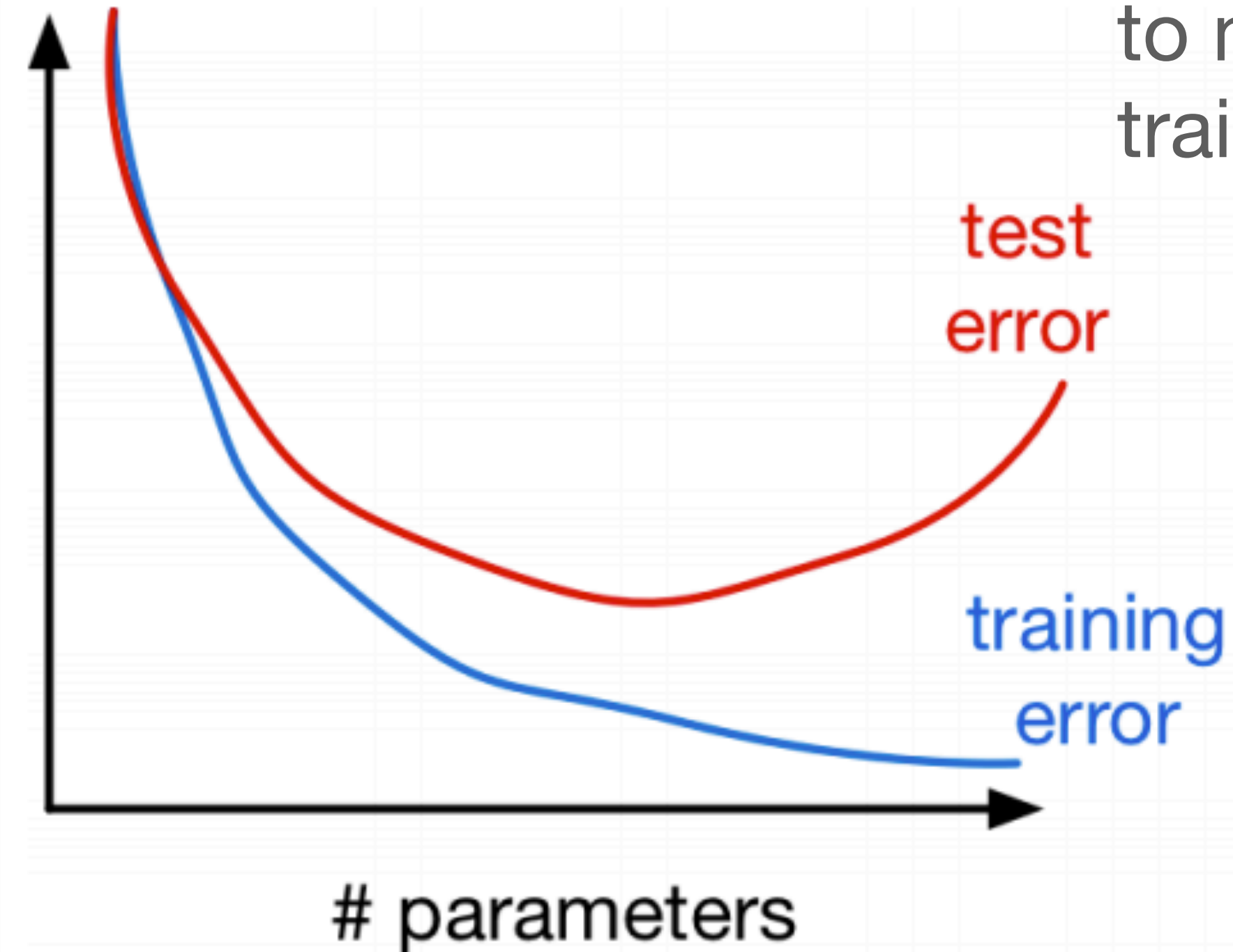
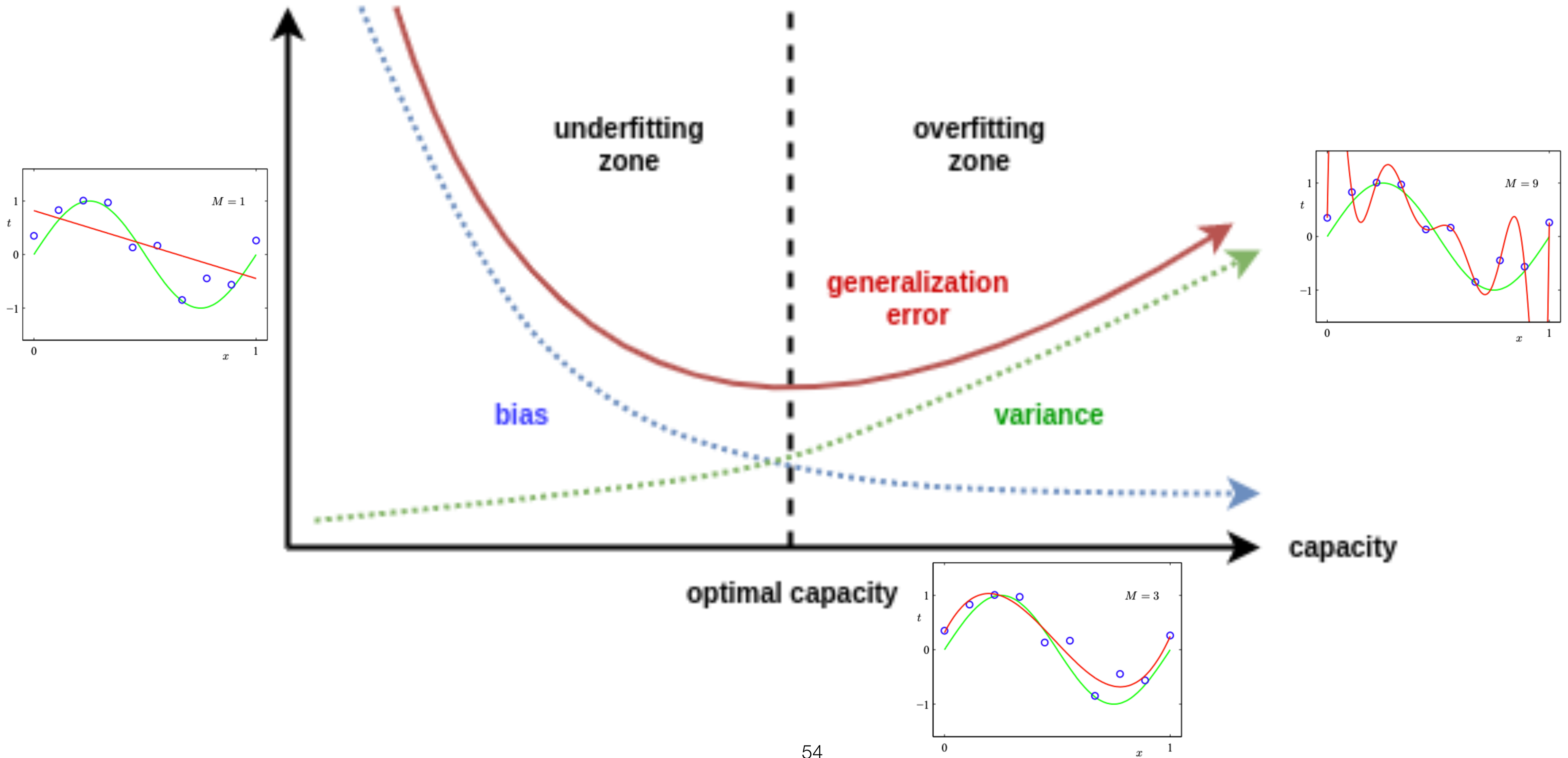


Image Credits: Roger Grosse

<https://csc413-2020.github.io/assets/readings/L07a.pdf>

Bias and variance tradeoff



Validation

Train

Validation

Test

- Choose a metric: Precision/Recall/F1
- Optimize for metric on **Validation** (aka Development) set
- Finally evaluate on ‘unseen’ **test** set
- Cross-validation:
 - Repeatedly sample several train-val splits
 - Reduces sampling bias due to sampling errors

Train Valid

Train Valid

⋮

Valid Train

Aggregating scores

- We have Precision, Recall, F1 for each class
- How to combine them for an overall score?
- **Macro-average**: Compute for each class, then average
- **Micro-average**: Collect predictions for all classes and jointly evaluate

Macro vs Micro average

Class 1

	Truth: yes	Truth: no
Classifier: yes	10	10
Classifier: no	10	970

Class 2

	Truth: yes	Truth: no
Classifier: yes	90	10
Classifier: no	10	890

Micro Ave. Table

	Truth: yes	Truth: no
Classifier: yes	100	20
Classifier: no	20	1860

Precision

- Macro-averaged precision: $(0.5 + 0.9) / 2 = 0.7$
- Micro-averaged precision: $100/120 = 0.85$
- Micro-averaged score is dominated by score on **common** classes

Summary

- Evaluation Metrics
 - **Accuracy** - coarse metric
 - **Precision**, **Recall**, **F1** for each class
- Aggregated scores
 - **Macro-average**: Compute for each class, then average
 - **Micro-average**: Collect predictions for all classes and jointly evaluate (dominated by common classes)
- Precision-Recall curve: pick threshold for maximum F1
- Use **validation** set to tune hyperparameters, **test** set should remain “unseen”